A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Energy conservation: transport, buildings, industry - Set 2

Mark scheme

AQA spec 3.3.4.2; Energy resources; textbook pp. 225-235

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Total = 42 + 26 + 31 + 9 + 22 = 130 kW **1** 31 / 130 x 100 = 23.8% (accept 24%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1 Cavity wall insulation:** the foam or fibre traps air in small pockets so convection in the cavity is prevented and the low thermal conductivity of still air reduces conduction through the wall **1 Low-emissivity double glazing:** the still air or inert gas between the panes conducts heat poorly, and the low-emissivity coating reflects infrared radiation back into the room instead of letting it radiate out **1 Heat exchanger:** outgoing stale warm air passes next to, but does not mix with, incoming cold fresh air, so most of the heat is transferred to the incoming air and the building can be ventilated without losing that heat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each extra layer saves less heat than the one before, because the layers already in place carry most of the temperature difference, so the saving per unit thickness falls **1** Beyond a certain thickness the cost and the embodied energy of making and fitting more insulation are never repaid by the energy it saves **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** A large power station runs at higher temperatures and pressures and uses large efficient turbines, so it converts a greater proportion of the fuel energy into electricity **1** The heat rejected by a large station is low-grade and the station is usually too far from any user, so it is lost to a river or cooling tower **1** A CHP unit is in the building that needs the heat, so the heat rejected by its generator is piped straight to the heating and hot water, and 70 - 80% of the fuel energy is used rather than about 35% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **combined heat and power** → using the waste heat from generation for heating	3

Do not write outside the box

Question	Marking guidance	Marks
01.5	**Any 5 of 6** 1 mark each for any 5 of the 6 points below, in any order, maximum 5. **1** Occupancy sensors and programmable thermostats so that lighting and heating run only where and when people are present **1** LED lighting, which converts a far greater proportion of the electricity into light and much less into waste heat than filament lamps **1** Large south-facing windows and a thermally massive structure to make use of passive solar gain, with shading to prevent overheating in summer **1** Heating water on demand at the point of use, so no energy is wasted keeping a tank hot and none is lost in the pipe run **1** Choosing materials with low embodied energy, and re-using the existing structure rather than demolishing and rebuilding **1 Changes in behaviour:** turning equipment off rather than leaving it on standby, closing doors, and setting the thermostat a degree lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-235; AQA spec 3.3.4.2.2-3.3.4.2.3.	

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